Manufacturing Industries

The big idea

Production of goods in large quantities after processing raw materials into more valuable products is called **manufacturing**. Paper from wood, sugar from sugarcane, iron and steel from iron ore, aluminium from bauxite: all are manufacturing. People in the secondary sector are engaged in manufacturing. A country's economic strength is measured by the development of its manufacturing industries.

Importance of manufacturing

- Helps **modernise agriculture**, the backbone of our economy.
- Reduces heavy dependence of people on agricultural income by providing **jobs in secondary and tertiary sectors**.
- Industrial development is a precondition for **eradicating unemployment and poverty**; this was the main philosophy behind public sector industries and joint ventures in India. It also aimed to bring down **regional disparities** by setting up industries in tribal and backward areas.
- **Export** of manufactured goods expands trade and commerce and brings foreign exchange.
- Countries that transform raw materials into a wide variety of higher-value finished goods are prosperous.

AGRICULTURE AND INDUSTRY GO HAND IN HAND

Agro-industries in India have given a major boost to agriculture by raising productivity. Industries depend on agriculture for raw materials (cotton, jute, sugarcane), and sell products like **irrigation pumps, fertilisers,**

insecticides, pesticides, plastic and PVC pipes, machines and tools to farmers. They are not exclusive of each other.

Contribution to national economy. Over the last two decades, the share of manufacturing in GDP has stagnated at about **17 per cent** (out of 27% for industry, including mining, quarrying, electricity and gas). The desired growth rate over the decade is **12 per cent**. Since 2003, manufacturing has grown at 9–10 per cent per annum. The government set up the **National Manufacturing Competitiveness Council (NMCC)** with the objective of improving productivity through appropriate policy interventions.

Industrial location

Industrial locations are complex: they are influenced by availability of **raw material, labour, capital, power and market**. It is rarely possible to find all these at one place, so manufacturing tends to locate at the most appropriate place where all factors can be made available or arranged at lower cost.

- Many industries tend to come together to make use of advantages offered by urban centres, known as **agglomeration economies**.
- After independence, most industries were concentrated in a few cities like Mumbai, Kolkata and Chennai; pre-independence industries were located from the point of view of overseas trade.

Classification of industries

Basis	Types	Examples
Source of raw material	Agro-based / Mineral-based	Cotton, woollen, jute, silk textile, rubber, sugar, tea, coffee, edible oil / iron and steel, cement, aluminium, machine tools, petrochemicals

Basis	Types	Examples
Prime role	Basic or key industries (supply products to manufacture other goods) / Consumer industries	Iron and steel, copper smelting, aluminium smelting / sugar, toothpaste, paper, sewing machines, fans
Capital investment	Small-scale (maximum investment allowed on assets of a unit is defined, e.g., up to ₹1 crore as in NCERT) / Large-scale	
Ownership	Public sector (government) / Private sector (individuals) / Joint sector (state and individuals) / Cooperative sector (producers or suppliers of raw materials, workers or both)	**BHEL, SAIL / TISCO, Bajaj Auto, Dabur / Oil India Ltd (OIL) / sugar industry in Maharashtra, coir industry in Kerala**
Bulk and weight of raw material and finished goods	Heavy / Light	Iron and steel / electrical goods

Agro-based industries

TEXTILE INDUSTRY

The only industry in India that is **self-reliant and complete in the value chain**, from raw material to the highest value-added products. It contributes significantly to industrial production, employment generation (about 35 million directly, as per NCERT) and foreign exchange earnings. **Cotton textiles**: in ancient India, cotton textiles were produced by hand spinning and handloom weaving. After the 18th century, power-looms came into use. The **first successful textile mill** was established in **Mumbai in 1854**.

Early years: cotton textile mills were concentrated in the cotton-growing belt of **Maharashtra and Gujarat**, due to availability of raw cotton, market, transport including accessible port facilities, labour, moist climate, etc. India has world-class production in **spinning**, but **weaving** supplies low-quality fabric because it cannot use much of the high-quality yarn produced in the country. Weaving is highly decentralised (handloom, power-loom, mills). India exports yarn to **Japan**; other importers include the USA, UK, Russia, France, East European countries, Nepal, Sri Lanka and Bangladesh. Problems: erratic power supply, outdated machinery, low output of labour, stiff competition with synthetic fibres.

JUTE TEXTILES

India is the **largest producer of raw jute and jute goods** and stands at **second place as an exporter after Bangladesh**. Most mills are located in **West Bengal**, mainly along the banks of the **Hugli river**, in a narrow belt. The **first jute mill** was set up near **Kolkata in 1859 at Rishra**. After Partition in 1947, jute mills remained in India but three-fourths of the jute-producing area went to Bangladesh (then East Pakistan).

Location factors in the Hugli basin: proximity of jute-producing areas, inexpensive water transport, good network of railways, roadways and waterways, abundant water for processing raw jute, cheap labour from West Bengal and adjoining states of Bihar, Odisha and UP, and Kolkata as a large urban centre providing banking, insurance and port facilities for export.

Challenges: stiff competition from synthetic substitutes and from Bangladesh, Brazil, the Philippines, Egypt and Thailand. The **National Jute Policy was formulated in 2005** to increase productivity, improve quality, ensure good prices to farmers and enhance yield per hectare. Growing global concern for environment-friendly, biodegradable materials has opened opportunities for jute products.

SUGAR INDUSTRY

India stands **second as a world producer of sugar** but occupies the first place in the production of **gur and khandsari**. The raw material is bulky and its sucrose content reduces in haulage, so mills are located close to sugarcane areas. Over 460 mills are spread over UP, Bihar, Maharashtra, Karnataka, Tamil Nadu, Andhra Pradesh, Gujarat, Punjab, Haryana and MP. Sixty per cent of mills are in UP and Bihar. The industry is **seasonal**, ideally suited to the **cooperative sector**. In recent years, mills have shifted and concentrated in **southern and western states, especially Maharashtra**, because cane there has a **higher sucrose content**, the **cooler climate** ensures a longer crushing season, and **cooperatives are more successful** there. Challenges: seasonal nature, old and inefficient methods, transport delays, need to maximise use of bagasse.

Mineral-based industries

IRON AND STEEL INDUSTRY

The iron and steel industry is the **basic industry**, since all other industries, heavy, medium and light, depend on it for their machinery. It is a **heavy industry** because all raw materials and finished goods are heavy and bulky, entailing heavy transportation costs. Iron ore, coking coal and limestone are required in the ratio of approximately **4:2:1**; some manganese is also needed to harden the steel.

India is an important iron and steel producing country; it ranks **second among world crude steel producers** (after China; as per NCERT 2019) but is also the largest producer of sponge iron. Despite this, per capita consumption of steel is low. Most public sector steel plants market their steel through the **Steel Authority of India Limited (SAIL)**, while TISCO markets through **Tata Steel**.

Chotanagpur plateau region has the maximum concentration of iron and steel industries, mainly because of the low cost of iron ore, high-grade raw materials in proximity, cheap labour and vast growth potential in the home market.

Problems: high costs and limited availability of coking coal, lower productivity of labour, irregular supply of energy and poor infrastructure. Liberalisation and FDI have boosted the industry.

ALUMINIUM SMELTING

The **second most important metallurgical industry** in India. Aluminium is light, resistant to corrosion, a good conductor of heat, malleable, and becomes strong when mixed with other metals. Used to manufacture aircraft, utensils and wires; it has gained popularity as a substitute for steel, copper, zinc and lead. Smelters are located in **Odisha, West Bengal, Kerala, Uttar Pradesh, Chhattisgarh, Maharashtra and Tamil Nadu**. **Bauxite**, the raw material, is a very bulky, dark reddish coloured rock. Regular supply of electricity and assured sources of raw material at minimum cost are the two prime factors for location.

Chemical industries: comprise both **large and small-scale** units; includes **inorganic** (sulphuric acid, nitric acid, alkalies, soda ash, caustic soda) and **organic** chemicals (petrochemicals for synthetic fibres, plastics, dyes, pharmaceuticals). Chemical industry is its own largest consumer.

Fertiliser industry: centred around the production of **nitrogenous fertilisers (mainly urea)**, phosphatic fertilisers and ammonium phosphate (DAP) and complex fertilisers. India is the **third largest producer** of nitrogenous fertilisers. **Gujarat, Tamil Nadu, Uttar Pradesh, Punjab and Kerala** contribute half of the fertiliser production. Expansion came after the Green Revolution.

Cement industry: essential for construction (houses, factories, bridges, roads, airports, dams). Requires bulky and heavy raw materials like **limestone, silica and gypsum**. The **first cement plant was set up in Chennai in 1904**. After independence the industry expanded; decontrol of price and distribution since 1989 and other policy reforms led to rapid growth. Plants are in Gujarat (with access to markets in the Gulf), etc.

Automobile industry: provides vehicles for quick transport of goods, services and passengers. After liberalisation, the coming of new and contemporary models stimulated demand. Located around Delhi, Gurugram, Mumbai, Pune, Chennai, Kolkata, Lucknow, Indore, Hyderabad, Jamshedpur and Bengaluru.

Information technology and electronics industry: covers a wide range of products from transistor sets to television, telephones, cellular telecom, computers. **Bengaluru** has emerged as the **electronic capital of India**. A major impact of this industry is **employment generation**, and it has been a major earner of foreign exchange through the fast-growing **Business Process Outsourcing (BPO)** sector. Continuing growth in hardware and software is key to India's success. A large number of women are employed in this sector.

Industrial pollution and environmental degradation

Industries are responsible for four types of pollution:

1. **Air pollution**: caused by high proportion of undesirable gases like **sulphur dioxide and carbon monoxide**; airborne particulate materials (solids and liquids) like dust, sprays, mist and smoke. Smoke is emitted by chemical and paper factories, brick kilns, refineries and smelting plants, and burning of fossil fuels. Toxic gas leaks can be very hazardous with long-term effects (e.g., **Bhopal gas tragedy, 1984**).
2. **Water pollution**: caused by **organic and inorganic industrial effluents** discharged into rivers. Main culprits: paper, pulp, chemical, textile and dyeing, petroleum refineries, tanneries and electroplating industries that let out dyes, detergents, acids, salts and heavy metals like lead and mercury, pesticides, fertilisers, synthetic chemicals with carbon, plastics and rubber. Fly ash, phospho-gypsum and iron and steel slags are major solid wastes.
3. **Thermal pollution**: occurs when **hot water from factories and thermal plants** is drained into rivers and ponds before cooling. Wastes from nuclear power plants and nuclear and weapon production facilities cause cancers, birth defects and miscarriages.

4. **Noise pollution**: results in irritation and anger, can cause hearing impairment, increased heart rate and blood pressure. Sources: industrial and construction activities, machinery, factory equipment, generators, saws, pneumatic and electric drills.

   Land and soil pollution: dumping of wastes (glass, harmful chemicals, industrial effluents, packaging, salts and garbage) makes soil useless; rainwater percolating carries pollutants to groundwater.

CONTROL OF ENVIRONMENTAL DEGRADATION

- Minimise use of water by **reusing and recycling** in two or more successive stages.
- **Harvest rainwater** to meet water requirements.
- **Treat hot water and effluents** before releasing into rivers and ponds. Treatment can be primary (mechanical: screening, grinding, flocculation and sedimentation), secondary (biological process) and tertiary (biological, chemical and physical processes, recycling of wastewater).
- Overdrawing of groundwater by industry must be regulated legally.
- Particulate matter in air can be reduced by fitting smoke stacks with **electrostatic precipitators, fabric filters, scrubbers and inertial separators**. Smoke can be reduced by using **oil or gas instead of coal**.
- Machinery can be redesigned for energy efficiency and reduced noise; use of noise-absorbing material and earplugs/earphones.

  Example: NTPC has shown the way by preserving the natural environment and resources (e.g., at **Ramagundam**): optimum utilisation of equipment through latest technologies, minimising waste generation by maximising **ash utilisation**, providing **green belts** for nurturing ecological balance, reducing environmental pollution through ash pond management, ash water recycling and liquid waste management, and **ecological monitoring**.

PUBLIC VS JOINT SECTOR

Oil India Ltd is a **joint sector** industry; **BHEL and SAIL** are **public sector**; **TISCO and Bajaj Auto** are **private sector**. Cooperative examples: **sugar in Maharashtra** and **coir in Kerala**.

BOARD FAVOURITES AND MAP WORK

"Importance of manufacturing", "Agriculture and industry are complementary: explain", "Why is the sugar industry shifting to southern and western states?", "Factors for location of jute mills along the Hugli", "Classify industries on the basis of ownership" and "How can industrial pollution be controlled?" (5 marks). **Map: Cotton textile industries**: Mumbai, Indore, Surat, Kanpur, Coimbatore. **Iron and steel plants**: Durgapur, Bokaro, Jamshedpur, Bhilai, Vijayanagar, Salem. **Software technology parks**: Noida, Gandhinagar, Mumbai, Pune, Hyderabad, Bengaluru, Chennai, Thiruvananthapuram.

Quick recap

- Manufacturing: processing raw materials into more valuable products; key to jobs and reducing poverty.
- Location factors: raw material, labour, capital, power, market; agglomeration economies.
- Classification by raw material, role, capital, ownership and bulk.
- First cotton mill Mumbai 1854; first jute mill Rishra 1859; National Jute Policy 2005; first cement plant Chennai 1904.
- Sugar shifting to Maharashtra and the south (higher sucrose, cooler climate, cooperatives).
- Iron and steel is a basic, heavy industry concentrated in Chotanagpur; aluminium is second most important metallurgical industry.
- Pollution: air, water, thermal, noise; NTPC as a model of environmental care.

Practice questions

1. Which of the following is a joint sector industry?
   (a) BHEL (b) TISCO (c) Oil India Ltd (d) Coir industry in Kerala
2. The first successful textile mill in India was established in
   (a) Ahmedabad, 1861 (b) Mumbai, 1854 (c) Kanpur, 1860 (d) Kolkata, 1859
3. Most jute mills in India are located along the
   (a) Godavari (b) Narmada (c) Mahanadi (d) Hugli
4. Why is the sugar industry shifting to southern and western states?
   (a) Cane there has higher sucrose content, the cooler climate gives a longer crushing season, and cooperatives are more successful (b) These states have the most rainfall in India (c) Sugar mills are banned in UP and Bihar (d) There is no market for sugar in the north
5. Which city is known as the electronic capital of India?
   (a) Hyderabad (b) Mumbai (c) Bengaluru (d) Chennai
6. Thermal pollution occurs when
   (a) smoke is emitted by brick kilns (b) loud machinery is used near houses (c) garbage is dumped on land (d) hot water from factories is drained into rivers before cooling
7. Aluminium smelting is the
   (a) most important agro-based industry (b) second most important metallurgical industry in India (c) only industry complete in its value chain (d) oldest industry of India
8. "Agriculture and industry are complementary to each other." Explain.
9. Suggest any five measures to control environmental degradation caused by industries.

Answers: 1. c 2. b 3. d 4. a 5. c 6. d 7. b 8. see notes 9. see notes